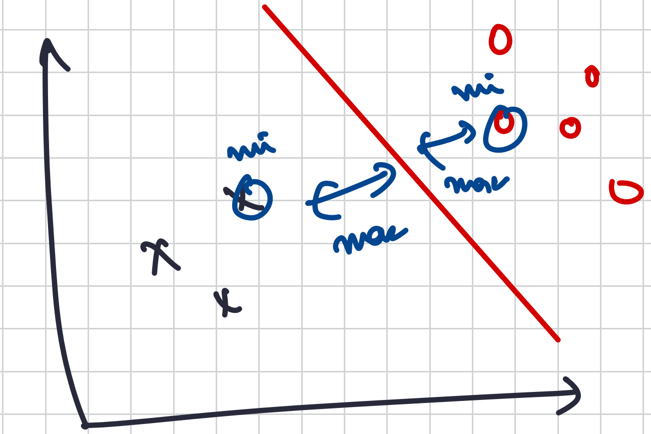
support vectors II

w, b



$$\alpha \left\{ \begin{array}{c} \min \quad \|w\|^2 \\ \text{s.t.} \\ y_i(w^T x + b) \geq 1 \end{array} \right\}$$

max.
penalty

primal $\min \max \{ \|w\|^2 - \alpha \{ (w^T x + b) y - 1 \} \}$

↓

dual

parent → save money

min.
shop

Kid → Toy
↓
most expensive toy in the shop...

max.
toy

	Toy1	Toy 2
Shop1	100	400
Shop2	300	200

$$\left\{ \underset{\text{Kid I}}{\max} \left\{ \overset{\text{P II}}{\min} \qquad \right\} \right.$$

$$\overset{\text{P I}}{\underset{\text{choose first}}{\min}} \left\{ \overset{\text{K II}}{\max} \left\{ \underset{\text{Kid decide}}{} \qquad \right\} \right\}$$

$$L(w,b,\alpha) = \frac{1}{2}\|w\|^2 - \sum_{i=1}^{n} \alpha_i (y_i (w \cdot x_i + b) - 1]$$

$\alpha_i \geq 0 \; \forall$

$$\left.\begin{array}{l} \frac{\partial L}{\partial w} = 0 \\ \frac{\partial L}{\partial b} = 0 \end{array}\right\}$$

α : Constant

max $\omega(\alpha) = \Sigma \alpha_i$

$-\frac{1}{2} \Sigma\Sigma y_i y_j \alpha_i \alpha_j$ $\boxed{K(\phi(x_i), \phi(x_j))}$

$\boxed{\text{scalar}}$

$$w^* = \Sigma \alpha_i y_i x_i$$

$$b^* = \sum_{i=1}^{n} \alpha_i y_i = 0$$

$$\text{max } W(\alpha) = \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \Sigma\Sigma \alpha_i \alpha_j y_i y_j \underline{(x_i \cdot x_j)}$$

dot product

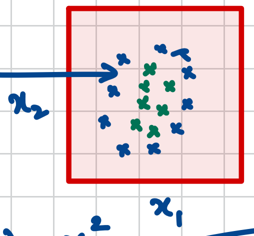
dot product :→ computed with some equivalent fxn

x_1, x_2 → $x_1 \cdot x_2$ → $\phi(x_1) \cdot \phi(x_2)$

~~x_1^2~~

~~x_2^2~~

$\boxed{1, x_1 x_2}$

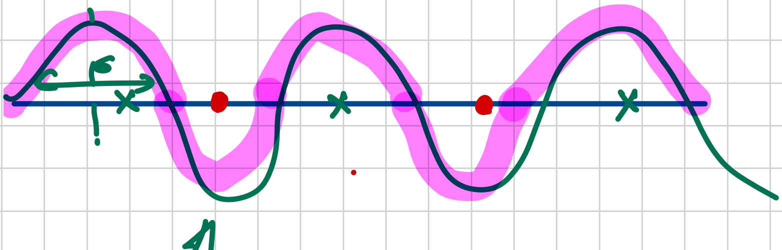


$\phi(\cdot) = x^2$

$$\underline{K(x_1, x_2)} \equiv \phi(x_1) \cdot \phi(x_2)$$

Gaussian!

↳ Kernel → Gaussian Kernel

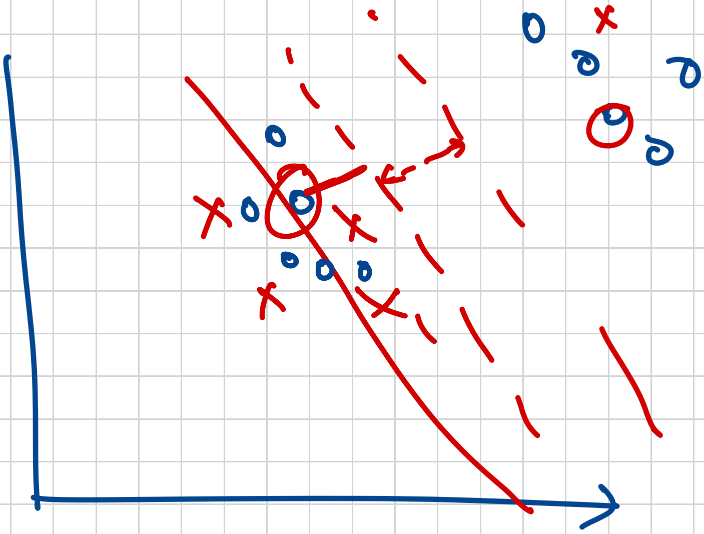


$$e^{-\frac{(x_i - x_j)}{2\sigma^2}}$$

Kernel trick!

distn

$$\| x_i - x_j \|^2$$



$$\max . \quad \frac{w^T x + b}{\| w \|}$$

s.t.

cost

$$\min \left\{ \| w \|^2 + C \xi_i^2 \right.$$

s.t.

$$\left\{ (y_i)(w^T x + b) \right\} \geq 1 - \xi_i$$